

Max Auer, PhD

Curriculum Vitae

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Research Interests

Dynamical Systems, Ergodic Theory, Probability Theory, Random Dynamical Systems, Limit Theorems, Rare Events and Extremes

Education

- Jan.2021–
May2025 **PhD in Mathematics**, *University of Maryland, College Park, Department of Mathematics*, Advisors: Dmitry Dolgopyat and Adam Kanigowski
- Oct.2018–
Aug.2020 **Master of Science in Mathematics**, *University of Vienna*, Graduated with distinction, Master Thesis: 'Local limit theorems for hitting and return times', Advisor: Roland Zweimüller, Specialisation: Probability Theory and Dynamical Systems
- Oct.2015–
Aug.2018 **Bachelor of Science in Mathematics**, *University of Vienna*, Graduated with distinction, Bachelor Thesis: 'Weak convergence of measures', Advisor: Roland Zweimüller

Research Positions

- Sept.2025–
present **Postdoctoral Research Assistant**, *University of Queensland*, Topic: 'Random Dynamical Systems', PI: Cecilia González-Tokman
- Jul.2025–
Aug.2025 **Postdoctoral Research Assistant**, *University of Vienna*, Topic: 'Rare events in Dynamical Systems', PI: Roland Zweimüller
- Jan.2023–
May 2025 **Research Assistant**, *University of Maryland, College Park*, Topic: 'Stochastic Properties of Dynamical Systems', PI: Dmitry Dolgopyat
- Aug.2020–
Jul.2021 **Research Assistant**, *University of Vienna*, Topic: 'Rare events in Dynamical Systems', PI: Roland Zweimüller

Publications

- 2026 **Quenched limit theorems via mixing of all orders for random dynamical systems**, *Max Auer*, arXiv:2605.30964, preprint
We adapt the notion of mixing of all orders to random dynamical systems and use it to establish quenched limit theorems.
- 2026 **Trimmed strong laws and distributional limits for exponentially mixing systems**, *Max Auer, S. Liu*, arXiv:2601.08126, preprint
We study ergodic sums of non-integrable observables for exponentially mixing systems.
- 2025 **Local limit theorems for hitting times and return times of small sets**, *Max Auer, R. Zweimüller*, arXiv:2312.14581, To appear in Israel Journal of Mathematics
We show a sharper version of the Poisson limit theorem in some deterministic systems.

- 2025 **Trimmed ergodic sums for non-integrable functions with power singularities over irrational rotations**, *Max Auer, T. I. Schindler*, arXiv:2503.22242, preprint
We study ergodic sums of non-integrable observables over rotations.
- 2023 **Poisson Limit Theorems for Systems with Product Structure**, *Max Auer*, *Discrete Contin. Dyn. Syst.* 45 (2025), no. 5, 1454–1491, We develop a method for showing the Poisson limit theorem in some deterministic systems with intermediate behaviour.

Selected Talks

- 2026 **Strong laws and distributional limits for trimmed ergodic sums**, *MATRIX Research Program: Statistical Properties and Extremes in Dynamical Systems Theory and Numerics*, Creswick, Australia
- 2024 **Poisson Limit Theorems for Systems with Product Structure**, *Conference: Probabilistic Techniques for Random and Time-Varying Dynamical Systems*, ESI, Vienna, Austria

Teaching and Supervision

- 2026–present **Honours Thesis Advisor**, *University of Queensland*, Tipping phenomena in dynamical systems
- Aug.2021–
Dec.2022 **Teaching Assistant**, *University of Maryland, College Park*, MATH240: Introduction to Linear Algebra
- Aug.2019–
Dec.2019 **Tutor**, *Faculty of Mathematics, University of Vienna*

Awards

- 2023 **Spotlight on Graduate Research Martin Monroe Gold Medal**, *University of Maryland, College Park, Department of Mathematics*
- 2021 **Prize in recognition of academic achievement**, *Würdigungspreis des Bundesministeriums für Bildung, Wissenschaft und Forschung*, Austria

Programming

Python, MATLAB, $\text{\LaTeX}$